

Subnetting Solution

Given

- ▶ IP Address: 192.168.100.0
- ▶ Subnet Mask: 255.255.255.0 (/24)
- ▶ Required Subnets: 4

Step 1: Find New Subnet Mask

We need 4 subnets.

$$2^n \geq 4 \Rightarrow n = 2 \text{ bits borrowed}$$

Original mask: /24

New mask:

$$/24 + 2 = /26$$

New subnet mask:

$$255.255.255.192$$

Block size:

$$256 - 192 = 64$$

Step 2: Subnet Creation

We divide 0–255 into blocks of 64.

Subnet 1

- ▶ Network Address: 192.168.100.0
- ▶ Broadcast Address: 192.168.100.63
- ▶ Host Range: 192.168.100.1 – 192.168.100.62

Subnet 2

- ▶ Network Address: 192.168.100.64
- ▶ Broadcast Address: 192.168.100.127
- ▶ Host Range: 192.168.100.65 – 192.168.100.126

Subnet 3

- ▶ Network Address: 192.168.100.128
- ▶ Broadcast Address: 192.168.100.191
- ▶ Host Range: 192.168.100.129 – 192.168.100.190

Subnet 4

- ▶ Network Address: 192.168.100.192
- ▶ Broadcast Address: 192.168.100.255
- ▶ Host Range: 192.168.100.193 – 192.168.100.254

Final Summary

Subnet	Network Address	Broadcast Address	Usable Hosts
1	192.168.100.0	192.168.100.63	62
2	192.168.100.64	192.168.100.127	62
3	192.168.100.128	192.168.100.191	62
4	192.168.100.192	192.168.100.255	62